ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/352490972

‘We live and die in chestnut’: remaining and adapting in the face of pest and disease outbreak in Turkey

Article · June 2021

DOI: 10.1080/01426397.2021.1929888

CITATIONS
2

READS
141

6 authors, including:



Jeffrey Robert Wall
University of Guelph
11 PUBLICATIONS **63** CITATIONS
SEE PROFILE



Elif Başak Aksoy
Hacettepe University
9 PUBLICATIONS **38** CITATIONS
SEE PROFILE



coşkun Köse
İstanbul University-Cerrahpaşa
52 PUBLICATIONS **352** CITATIONS
SEE PROFILE

All content following this page was uploaded by Taner Okan on 05 July 2021.

The user has requested enhancement of the downloaded file.



Landscape Research

Routledge
Taylor & Francis Group

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/clar20

‘We live and die in chestnut’: remaining and adapting in the face of pest and disease outbreak in Turkey

Jeffrey Robert Wall, Nesibe Köse, Elif Başak Aksoy, Coşkun Köse, Taner Okan & Shorna Allred

To cite this article: Jeffrey Robert Wall, Nesibe Köse, Elif Başak Aksoy, Coşkun Köse, Taner Okan & Shorna Allred (2021): ‘We live and die in chestnut’: remaining and adapting in the face of pest and disease outbreak in Turkey, Landscape Research, DOI: 10.1080/01426397.2021.1929888

To link to this article: https://doi.org/10.1080/01426397.2021.1929888

Published online: 17 Jun 2021.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=clar20

LANDSCAPE RESEARCH
https://doi.org/10.1080/01426397.2021.1929888

landscape research group

Routledge
Taylor & Francis Group

Check for updates

'We live and die in chestnut': remaining and adapting in the face of pest and disease outbreak in Turkey

Jeffrey Robert Wall [a,b], Nesibe Köse [c], Elif Başak Aksoy [d], Coşkun Köse [e], Taner Okan [f] and Shorna Allred [g]

[a]Department of Geography, Environment and Geomatics, University of Guelph, Guelph, Canada; [b]Wildlife Conservation Society of Canada, Toronto, Canada; [c]Department of Forest Botany, Faculty of Forestry, Istanbul University-Cerrahpaşa, Istanbul, Turkey; [d]Department of Anthropology, Hacettepe University, Ankara, Turkey; [e]Department of Forest Biology and Wood Protection Technology, Faculty of Forestry, Istanbul University-Cerrahpaşa, Istanbul, Turkey; [f]Forestry Economics Department, Faculty of Forestry, Istanbul University-Cerrahpaşa, Istanbul, Turkey; [g]Department of Natural Resources, Cornell University, Ithaca, USA

ABSTRACT

Concern is growing worldwide over the negative outcomes of rural abandonment. Yet, problematisation of this phenomenon remains limited by insufficient explanatory frameworks and lack of empirical evidence from the conditions which precede, underlie and succeed it. Accordingly, this paper presents a case from Turkey, where significant rural abandonment is locally attributed to the ravages of multiple introduced pathogens in European chestnut (*Castanea sativa*) populations, and where our previous investigation has verified that traditional livelihood practices mitigate damage severity at the levels of trees, plots and landscapes. In order to better understand individual stakeholder motivations for remaining acting members of chestnut landscapes in the face of such serious challenges, we conducted 142 extended ethnographic and narrative interviews with chestnut-utilising participants across Turkey's highly diverse human and physical geography. Our results show how the struggle to remain as acting landscape members requires community livelihood adaptation, drawing on institutional memory, innovative learning and social connectedness.

KEYWORDS

Castanea sativa; community adaptation; cultural resilience; rural abandonment; traditional livelihood practices

Introduction

There is a popular local theory from Turkey's Black Sea Province of Sinop, where chestnuts are endemic and historic (Mattioni et al., 2010; Topcu et al., 2009). The theory states that the onset of chestnut blight in the 1980s and 1990s ignited the region's well-known rural exodus (FAO, 2009; Okan et al., 2017). In fact, the entire chestnut population of Turkey, distributed across Turkey's Black Sea, Marmara and Aegean regions, endures the compounding effects of three exotic pathogen outbreaks introduced through trade over the last century. These include ink disease caused by *Phytophthora cinnamomi* (Erdem, 1951), followed by the chestnut blight caused by *Cryphonectria parasitica* (Akdogan & Erkman, 1968) and most recently, the gall wasp, *Dryocosmus kuriphilus* (Cetin et al., 2014). At a moment when globalised acceleration of the spread and severity of forest pathogens continues unabated (Ghelardini et al., 2016), the cases of agrarian communities like those most engaged with the chestnut tree in Turkey can help to shed light on the social dimensions of this issue as it relates to rural landscape abandonment. We take the local theory of Sinop's rural

CONTACT Coşkun Köse ckose@istanbul.edu.tr

© 2021 Landscape Research Group Ltd

abandonment to be compelling evidence for the need to improve upon prevailing theories and anaemic empirical knowledge of rural abandonment so that they can more fully account for the conditions which precede, underlie and succeed the phenomenon of communities leaving the geographies to which they know they belong.

Rural abandonment has historically enjoyed widespread, though uncritical, support by international and national governance structures (Davis, 2006; Hobbs & Cramer, 2007; McMichael, 2008) based on select—and increasingly dated—ideas from economics and ecology (Lewis, 1954; Luken, 1990; Mather, 1992; Navarro & Pereira, 2015; Rostow, 1960). There is growing recognition, however, that this rapidly expanding category of landcover change (Hobbs & Cramer, 2007) depletes the knowledge and traditions of rural ecological maintenance (Berkes et al., 2000; Rapport & Maffi, 2010; Voeks, 2007) *and* leads to numerous unfavourable environmental outcomes (Otero et al., 2015; Plieninger et al., 2014; Queiroz et al., 2014). Two important fronts in the emerging problematisation of rural abandonment are enriching its explanatory frameworks and building empirical evidence of the socio-ecological conditions which precede, underlie and succeed it.

While economics is the disciplinary basis of prevailing explanations of rural abandonment, most economic studies do not consider on-the-ground evidence, instead favouring aggregated official statistics (see Liao et al., 2017; Parry et al., 2010). By contrast, ecology has compiled much empirical evidence from contexts of abandoned land (Hobbs & Cramer, 2007; Luken, 1990; Mather, 1992) but has by and large waived the opportunity to contribute to explanatory frameworks of rural abandonment. Despite this abdication, ecology has a major contribution to make along with the social sciences to an explanatory framework of rural abandonment. Hobbs and Cramer (2007) outline two simple and functional ecological hypotheses for the genesis of rural abandonment. First, overly exploitative local production practices can lead to unacceptable environmental states, such as soil exhaustion and overgrazed rangelands which in turn lead to abandonment. Second, non-local, ubiquitous economic and industrial externalities, such as air and water pollution, ubiquitous climate change, invasive species, and exotic pathogens, could be jeopardising local ecologically dependent livelihoods (Hobbs & Cramer, 2007).

The case of chestnut-utilising communities adapting to pest and disease pressure in Turkey represents a promising context to test prevailing explanatory frameworks for rural abandonment and for garnering evidence of the phenomenon in a context in which it is widely recognised to be happening. This is true for three reasons. First, the Sinop hypothesis itself clearly suggests significant local awareness that ecological instability is playing a role in the well-known rural exodus from the Black Sea region (Bugday & Özden, 2017) above and beyond the well-known economic incentives of economic out-migration. This improves upon the common economic theories on the issue while also highlighting the insufficiency of a threshold commonly adhered to in ecological studies by which ecological changes are attributed to human population fluctuation with no further inquiry made into the causes of this factor (Sanz et al., 2013). Second, it is well-established that chestnut-dominated forests have fared poorly in the wake of rural depopulation in Turkey (Akilli et al., 2009; FAO, 2009) and elsewhere (Zlatanov et al., 2013), a fact which highlights widespread under-consideration of the chronology of abandonment and its ecological implications. Third and finally, across Turkey there is significant variation in the key factors of geography, demography, culture and pest and disease severity timelines and investigating the phenomenon amidst such diversity promises to provide meaningful comparison between factors. This engagement with complexity lines up with priorities outlined by Aumeeruddy-Thomas et al. (2012) in France's Cévennes region, whose research with chestnut-engaged communities showcases the critical need to understand collective community navigation necessary to survive in the face of compounding challenges to cultural continuity.

Beyond explanatory capability and contextual information, an impactful problematisation engages 'the practical horizon of intelligibility within which problems come to matter for people' (Warner, 2002), and the more people the better. As such, the larger societal relevance of a problem of rural abandonment, were one to be spotlighted through evidence and explanation of evidence, would need to gain traction in explaining greater impact. In this paper, based on extensive evidence that 1) increased

individual engagement with chestnut trees significantly mitigates disease and pest pressure on individual trees (Wall et al., 2019), and, . 2) local capabilities of comprehending and strategising around complex ecological turbulence is highly effective (Okan et al., 2021), we continue the work of problematising rural abandonment by showing a methodology to perceive and support the force which sustains the struggle to remain: community livelihood adaptation. This proposed term is embedded within the theory and, importantly, the empirical methodology of community resilience.

We take community resilience to be a social group's ability to 'absorb a disturbance while retaining its essential functions' (Longstaff et al., 2010, p. 4). Perhaps due to their focus on natural disaster risk reduction Longstaff et al. (2010) have made important headway in developing a framework for 'community resilience' that accounts for the ecological resource base. They argue that community resilience is empirically observable in a community's adaptive capacity, the key attributes of which are 'institutional memory', 'innovative learning' and 'social connectedness' (Longstaff et al., 2010, p. 22) and that these are to be found binding to, among other things, features of the natural environment. In response to clear resonance with our participant testimony, we adopt these themes as key indicators of the adaptive capacity of communities in our research context.

This paper problematises widespread abandonment of cultural landscapes by engaging local motivations among concerned rural stakeholders across Turkey to remain in place in the face of mounting pest and disease pressure on European chestnut trees (*Castanea sativa*). We interweave previously established ecological evidence of landscape maintenance in the face of these challenges with ethnographic findings collected in the real-time context of community adaptation to three compounded disease epidemics for a cultural keystone species. Consequently, this study aims to contribute to the edge of multi-disciplinary investigations of rural abandonment by probing and describing its opposite, community resilience and adaptive capacity in a timely and illuminating research context. Further, we argue that this study begins the essential work of expanding upon prevailing explanatory frameworks for rural abandonment and showcasing a method for gathering empirical evidence from the socio-ecological conditions which precede, underlie and succeed it.

Materials and methods

Study context

In Turkey, the chestnut species is distributed across 262,000 hectares of the Eastern Black Sea, Western Black Sea, Marmara and Aegean provinces (Figure 1) and occurs across a wide variety of elevations, precipitation levels and temperature regimes (TNGDF, 2013). In the Eastern Black Sea, chestnut is found primarily in mixed-forest in high year-round precipitation, in steep topographic zones above 400 metres above sea level (MASL). In the Western Black Sea region, the terrain is relatively less mountainous, naturalised chestnut groves are present in state-governed forest between 20 and 600 MASL. The chestnut population in the arid Mediterranean climate of western Marmara is mainly found in state governed land at elevations higher than 600 MASL. In eastern Marmara, the more intensive horticultural technique of grafting is practiced. Grafting is the act of splicing a favoured plant cutting onto an already-grown specimen. Orchards of chestnut are present along the southern coast of the Marmara Sea at elevations as low as 20 MASL. Finally, the warm dry Mediterranean climates of the Aegean region host the most substantial grafted chestnut production in Turkey, all above 800 MASL.

Field work

Research was conducted during the summers of 2015 and 2016 by an interdisciplinary team of university-based scientists. In all, 10 provinces were selected that represent the geographic distribution and diversity of habitat for chestnut. In each province, 8–10 households were selected using purposive sampling (Miles & Huberman, 1984) to identify households known to engage in chestnut-related livelihood activities. This process was greatly assisted by local forestry officers, followed by

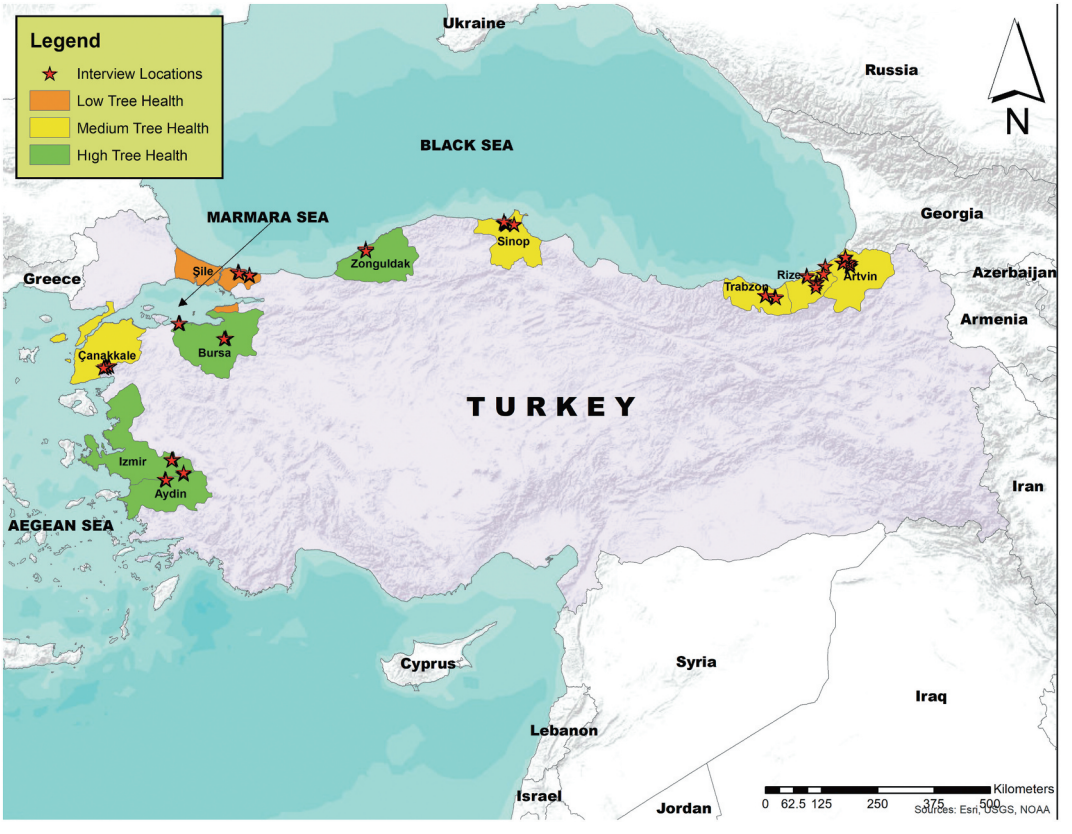


Figure 1. Map of Turkey and interview sites. Disease severity grouping of sites indicated by colour.

locally elected village representatives, known as *muhtars*. As part of our oral informed consent process, households were assured that participation was entirely anonymous and that our study took no stance on the validity, efficacy, or environmental beneficence of the livelihood practices that would be discussed. To account for gendered knowledge, in five of the 10 provinces, female household members were interviewed in privacy by someone of the same gender. All present household members were invited to participate in discussions. This was true in gender-segregated interviews as well, with only like-gendered household members participating. Adapted to cover the range of perspectives on the practice of livelihood, interviews were conducted to elicit: 1) narratives of livelihood history; 2) narratives of disease and pest onset; 3) strategies for adaptation and 4) prognoses for chestnut livelihood continuation. Interviews were conducted in Turkish and recorded by hand before being transcribed and coded off-site.

Data analysis

For qualitative analysis, interviewee transcripts were coded by the indicators of adaptive capacity: 'institutional memory', 'innovative learning' and 'social connectedness' (Longstaff et al., 2010). Special attention was paid to documenting report salience, when reports were repeated by households within and between sites. We took this salience to indicate institutional, or systemic, knowledge. In the process of coding, we sought to identify sub-categories which would help us to perceive the contours of and dynamics between 'institutional memory', 'innovative learning' and 'social connectedness.' Our final step in analysis was derived from our previous study in which we

determined that our research sites fell into three disease severity groups: low, medium and high (Identified in Figure 1). Accounting for the previous determination that the severity of disease and pest damage on trees was negatively correlated with the intensity of human involvement with trees and forested space, these groupings provide an illuminating scaffold for presenting interviewee perspective on their community capacity to perform traditional maintenance.

Results

One hundred and forty-two interviews were conducted with the members of 96 households in 10 provinces. At the outset of presenting results, it is important to describe the evolution and emergence of our final interviewee report categories. In addition to 'institutional memory', 'innovative learning' and 'social connectedness,' we determined two prominent sub-categories of institutional memory: epidemiological and narrative. We identified three prominent sub-categories of connectedness: local, non-local Formal and non-local informal. While examples illustrating the distinction between these sub-categories can be found in Table 1, 2 We took interviewee accounts of their observations, a description of the process of our discerning these categories warranted.

When asked to discuss the local history of chestnut production and/or collection, a significant number of participants began by discussing their impressions and recollection of the pests and diseases which have come to affect their chestnut trees over time. We categorised these as 'epidemiological' due to the distinct, scientific style in which they were delivered. Other chestnut-associated memories were delivered in more conversational accounts; these we categorised as 'narrative'.connectedness We took interviewee accounts of their observations, diagnoses and experiments applied in order to overcome chestnut pest and disease severity to be accounts of their innovative learning. We found no cause to identify sub-categories of innovative learning reports.

Social connectedness, on the other hand, turned out to be an unexpectedly rich, detailed subject in which interviewees had a lot to communicate. We accordingly developed categories to help with interpreting this body of reports. Local social connectedness, for instance, refers to salient statements referring to the importance of local social connectivity in the locally perceived experience of pest and disease pressure and the local response to it. Generational connectedness—or the connectedness of the old and young—was also routinely referenced by participants. Social connectedness with larger regional and even national social networks was routinely referenced by participants. We distinguished this field of connectivity as non-local. We discerned two main categories in non-local social connectedness: informal and formal. Informal non-local connectedness is made up of skills, information and resources exchanged with outside networks through voluntary, even organic outreach between like communities. For example, access to grafting service providers was routinely cited in some sites. These highly sought individuals acquire and distribute premier grafting materials along with knowledge of ecological suitability and perform expert grafting service.

Non-local formal connectedness referred to local community assessment of their influence with larger non-local institutions that were locally influential. This form of connectedness was primarily represented in a community's level of influence with the local GDF office. Mentions of the

Table 1. Categorisation of participant testimony by adaptive capacity category with general description.

Adaptive Capacity Category	Sub-Category	General Description
Institutional Memory	Epidemiological	Cause, dynamics and bio-physical impact of diseases
	Narrative	Cultural, economic, agronomic, horticultural chronology
Innovative Learning	————	Observations, diagnoses and experiments in agronomy, horticulture, ecology, economics and government relations
	Local	Identity, Generational
Social Connectedness	Non-local Informal	Locally accessible and exchangeable information, skills, resources, technology
	Non-local Formal	Influence over on-the-ground Forest Policy

Table 2. Adaptive capacity categories with sample testimony.

Category	Sample Testimony and Site
Epidemiological Memory	*'Mines, tree cutting, then disease.'* *'Government grafting programmes brought the disease.'* *'I have been collecting chestnuts for 31 years and have noticed disease for 5.'* *'At first, only grafted trees were sick, but then so were the wild trees.'*
Narrative Memory	*'Everyone used to have roasted chestnuts in their house for guests, but now they don't.'* *'Children used to go house to house collecting chestnuts from the neigbours.'* *'Other children and I used to go to school with our pockets stuffed with roasted chestnuts.'* *"We used to have black chestnut, which was tastier and survived better with disease, but now we don't."*
Innovative Learning	*'Grafted trees died quickly from blight.'* *"When we began grafting, for 20–30 years we used our own variety, then switched to Bursa. Surveying results by low, medium and high tree health sites"* *"Grafted ones are smart, wild ones are silly" (rhyming phrase in Turkish).* *'48 years ago we abandoned grape production for kestane b/c molasses price went too far down.'*
Local Social Connectedness	*'I use an old Circassian recipe that other villagers do not know.'* *"Labourers used to come from Inegol but now they're Kurdish."* *"There is a problem with theft by bears."* *"We are Yoruks so we use chestnut."* *"Everything from old times is made of chestnut, it is not good for young folks like us these days."* *"Chestnut is our house in life and in death."*
Non-Local Informal Social Connectedness	*'Yeah we grafters have a facebook page.'* *'My husband had a heart attack from the stress of the chestnut wholesale business.'* *"Chestnut work is so hard that there are no happy songs to sing while we're working."*
Non-Local Formal Social Connectedness	*"We are new to these varietal distribution programmes because we did not take advantage when the gov't first initiated the program 28 years ago."* *"Bursa Uludağ University is coming out with new blight resistant varieties."* *'Buyers used to come to village and now they don't because our harvest is not worth their time.'*

importance of this influence are prominent throughout the country, as almost all land reported to be used for chestnut collection and cultivation was state land.

Surveying results by low, medium and high tree health sites

In confluence with testimonial evidence of adaptive capacity across Turkey, geographic patterns of tree health documented in our previous study augment observations of the workings of adaptive capacity. This is because, as our previous study has shown (Wall et al., 2019), physical documentation and participant reporting of stakeholder engagement with trees and forests is highly correlated with increased tree health regardless of disease presence. Consequently, patterns to tree health can be safely understood as related to motivations and subjective accounts of participants and so also as an appropriate instrument for understanding them.

Low tree health

The lowest tree health group included the single site of Şile. This province is not far from the outer reaches of the metropolis of Istanbul. Industrial-scale resource extraction activities are evident throughout the province and two of these, mining and logging, were reported by interviewees to have had a significant effect on chestnut-related livelihood practices. In this site, we observed memories to be plentiful and detailed. Epidemiological memories, or diagnoses of low tree health, were particularly uniform and succinct across interviews here. Participants saw an iron correlation between the advent of mining with the advent of disease. 'Mining hurt the chestnut trees' (*Madencilik kestane ağaçlarına zarar verdi*), stated one participant. Another succinctly elaborated, 'mines, tree cutting, then disease' (*Madencilik, ağaç kesimi ve daha sonra hastalık*). At the same time, in Şile, interviewees recounted rich details from the past to corroborate the former abundance and

importance of chestnuts. They recalled numerous chestnut-affiliated traditions which are now gone including the local chestnut bazaar and a trick-or-treating like event for children with cheerful demands for roasted chestnuts from neighbours.

We observed a single, but notable, example of learning and adaptation in Şile. In our field excursions, we observed that a very high proportion of trees were coppiced in the area, meaning the main trunks of trees had been culled, allowing for clusters of smaller stems to eventually replace them. When we asked participants about this, we learned that it was a clear indication of the increasing favour of honey collection compared to nut collection. Participants explained that coppicing maximised the immediately available chestnut tree flowering as a forage for bees. That this certainly led to a decrease in nut harvest was well understood, but participants saw the overall transition as acceptable considering the already plummeting nut harvests due to mining, aggressive logging and disease.

It was clearly conveyed in Şile that weak non-local connectedness was integral to the calamitous state of local chestnut tree health. Ultimately, locals here repeatedly communicated a feeling of near powerlessness in the face of larger societal forces, specifically those that mount and legitimise local forest clear-cutting efforts and aggressive mining operations. A salient report in this tree health group is that local relations with local forest authority (under the General Directorate of Forestry (GDF)) are unsatisfactory. We observed an explanatory—and jarring—locally salient report that weak local influence over local environmental and forestry policy was in part due to the lack of solidarity of local demographics. In particular, it was argued that local identity diversity diminished solidarity and trust within the local population. Residents in Şile with whom we spoke viewed themselves as mixed communities and subsequently felt that they struggled to speak to local forest bureaucracy with one voice. By their reckoning, this resulted in unsatisfactory relations with local forest authority. Perhaps due to its proximity to the bureaucratic centre of Istanbul, Şile was affected by numerous state programs to resettle Turkish populations returning from former Ottoman territories, such as Bosnia and Circassia. Interviewees reported that this diversity was a key factor in aggressive state programs to clear-cut forest and implement mining in their vicinity.

Medium tree health group

The medium tree health group includes sites where intermediate tree health was observed. This group includes the eastern Black Sea sites in Sinop, Rize, Trabzon and Artvin as well as the site in western Turkey known as Çanakkale. On the ground, intermediate tree health translated to notable disease and pest damage, resulting in increasing local anxiety in combination with a rigorous persistence in chestnut livelihood activities. This dogged engagement is perhaps the most revealing contrast with the low tree health group, where disengagement was evident. Questions about chestnut-associated memories other than the advent of pest and disease pressure were received as questions into the truly mundane. As one participant stated from his porch in Rize, 'Sure, this whole house is made from chestnut. So is every other house in this village' (*Bu evin tamamı kestane ağacından yaplmış. Bu köydeki diğer evler de öyle*). Epidemiological memories, on the other hand were lucid and passionately reported. Participants routinely implicated the GDF in the compromised health of chestnut tree populations. A salient report in the province of Trabzon even declared that the chestnut blight arrived in the wake of a GDF-led grafting trial in the area.

Reports of learning were substantial in the medium tree health group. For instance, multiple reports identified the cause of declining chestnut tree health as the significant decrease in the number of people available to maintain chestnut groves by pruning trees, clearing brush and dead limbs. Participants decried that economic out-migration has led to too much shade, too much understory and too much leaf litter. One interviewee expressed this notion by arguing that failure to remove chestnut leaf litter led to unfavourable ecological conditions in chestnut-dominated forest. As he stated, 'Everyone knows that chestnut leaves have poison' (*Herkes biliyor ki kestane yapraklarında zehir var*). These reports show awareness of the local ecological aftermath of rural abandonment, which translates locally as a shortage of vigorous labourers and the insidious spread of undesirable landscapes.

A most salient cause cited for declining tree health was the restriction and prohibition by the GDF of traditional forest and orchard maintenance activities. The specific activities which participants argued were both restricted and ecologically beneficial included pruning, coppicing, prescribed burning, grazing of animals and the collection of waste wood for firewood and charcoal making. A vivid example of this ubiquitous sentiment was found in the western site of Çanakkale, where numerous participants made known that though grafting was prohibited on most local chestnut trees because they were on state land, favour for grafted trees was palpable. This local attitude was expressed in numerous references to grafted trees as 'smart' (*akıllı*) and un-grafted trees as 'insane' (*deli*).

High tree health group

The highest tree health group consisted of sites from an array of geographic zones including the Black Sea site of Zonguldak, the Marmara site of Bursa and the Aegean sites of Izmir and Aydın in Turkey's southwest. Across these sites, epidemiological memories were limited to estimations of lost harvest. However, an unexpectedly salient narrative emerged in local history, namely numerous women participants bemoaned the difficult transition that came with recent intensification of chestnut production. A woman in Aydın reported that the death of her husband came about due to stress related to the chestnut harvest period. Several interviewees missed certain cultivars of chestnut which had been abandoned in favour of meeting the demands of wholesale purchasers. These were missed for their flavours, aromas, peeling characteristics and more. In this same vein, a mother of a young bride-to-be in Bursa told how, in marriage negotiations with the groom's family on behalf of her daughter, she fought for and won her daughter the guaranteed right to be excluded from chestnut labour.

A very clear picture of innovative learning and its ability to flourish into non-local formal connectedness emerges from western Turkey. In Bursa, the majority of trees are grafted and located on villager-managed state land. Here, several interviewees reported on their recent efforts to engage with local government and university programs to acquire elite grafting materials including European-Japanese blight resistant cultivars. Similarly, the enormous transfer of production power from Bursa to Aydın and Izmir represents a noteworthy illustration of non-local formal connectedness. Once the undisputed capital of chestnut cultivation in Turkey, Bursa today produces just 5% of total national production today (TNGDF, 2013) due to pest and disease damage. Today, the provinces of Aydın and Izmir combined produce nearly 48% of Turkey's chestnuts (TNGDF, 2013). This seizure of new production possibility has been facilitated by the innovation of a unique policy arrangement with the GDF: the 49-year leases (*kırk dokuzluk*) derived from Forest Law 6831, whereby local village residents act as *de facto* managers and growers on state designated land. These lease innovations and this production can be understood as an expression of non-local formal connectedness.

Zonguldak, where tree health was the highest of the Black Sea sites, provides a unique illustration of the importance of local and non-local connectedness. In combination with its especially high tree health, Zonguldak stands out for its notable influence on local policy. This influence is embodied in the institution of the *sayvan*, a local term for a harvest hut—many capable of being locked—constructed and used at the site of collection, the forest plot. Of all the Black Sea sites, we only observed this tradition in Zonguldak. Two important observations can be made of this fact. First, outside of the Aegean region where chestnut was established and cultivated on leased land, the *sayvan* was the most overt sign of local harvest claim that we witnessed. It was reported to us that the builder of the hut and the keeper of the lock and key on each hut had a special entitlement to that space and to chestnuts collected in the vicinity. This speaks to a local collective agreement that demonstrates local connectedness.

Second, the very existence and persistence of these structures signifies a standing agreement with local GDF authorities, and one in which the position of the local community was unusually strong. As a testament to this fact, one participant boldly, but jokingly, declared that if a stranger came wandering in the forest to collect in their plot, they would be struck in the head. Such a statement was an absolute oddity in the region where the universal refrain was 'of course, anyone at all may collect here.' Villagers in Zonguldak exhibited a strong consensus to explain this evident strength of their position vis-à-vis local authorities. They pointed to the local economic lifeline that

was coal mining. Despite the fact that the well-known occurrence of coal mining was universally known to be responsible for Zonguldak chestnuts receiving the lowest price of any other province in regional markets, locals spoke reverentially of this employment resource, not least of which the numerous working age male heads of household who participated in our interview, a vivid anomaly for the Black Sea region. In sum, the overall conditions in Zonguldak, including economic sustenance, the presence of youth and particularly young males, healthy chestnut trees, clear and public rules for chestnut access, and self-perceived and expressed strength in relations with local authorities can best be understood as hanging together to support a low probability of rural abandonment from these villages for the time being, despite.

Discussion and conclusions

Our findings lend themselves especially well towards problematising rural abandonment in areas of Turkey where chestnut is an especially valued species. In place of an ambivalent acceptance that economic formulae—perhaps augmented by plant disease dynamics—drove the rural exodus of numerous otherwise unmoored individuals, it is clear from our engagement with stakeholders that a more complex hypothesis is necessary. We hypothesise that the rapid decrease in nut harvest which followed the spread of chestnut blight in Turkey—which was in fact the second major disease epidemic on chestnut trees introduced through trade in recent history—converged with other highly variable factors, such as the evolution of forest access restrictions, to further compromise the already tenuous basis for cultural-ecological continuity, thereby leading to an extreme demographic episode of out-migration and, iteratively, further jeopardy of chestnut-dominated forests.

This more robust perspective of an episode of rural abandonment has implications both in the research context and beyond. Our investigation of community adaptive capacity in our research context ended with a spotlight pointed at forest bureaucracy. Because we have documented both favourable and unfavourable local impressions of the activities enacted by forest bureaucrats and their offices and agencies, it is the complex truth that our findings offer this both critical and positive feedback. The most salient evidence of this as discussed by participants was also a factor which overdetermined the tree health grouping: variation in regulation *as practiced on the ground* by local GDF offices. Stated more directly, higher liberty to conduct traditionally obligatory tree and landscape management was a premier factor in individual explanations of successful tree and landscape maintenance. In the language of our study, this manifested in higher versus lower non-local formal institutional connectedness with local GDF offices and their practices. We believe our results provide grounds for the GDF to critically evaluate its routine disparagement of traditional village engagement with chestnut tree populations as disorderly forest use (*düzensiz faydalanmalar*) (TNGDF, 2013, pp. 7, 11, 18, p. 22). This traditional hands-on engagement and the deeply rooted motivation for it are best viewed as critical local investment in forest conservation (Pezzi et al., 2017). On the other hand, the GDF approaches, such as those observed in Aydın and Izmir which allow local branch offices the flexibility necessary to support local traditional maintenance, are worthy of emulation in other parts of the country.

More broadly, this study showcases the need for further inquiry into rural abandonment as a globally expanding category of landcover change, and how such inquiry can and should engage the human ecological dimensions. The present study provides insights for developing rural abandonment research at larger and smaller scales of analysis. At the mile-high level, there is analytical power to be gained by dissolving the unnecessary barriers between factors which contribute to rural abandonment (Aumeeruddy-Thomas et al., 2012). These include rural-urban migration, the growth of slums, environmentally irresponsible industrial practices in rural areas, climate change. Furthermore, there is a necessity to acknowledge the deep time horizon for all of these and chestnut landscapes, in particular, embody this fact (Michon, 2011). The perceptive power of considering various phenomena in this light is exemplified in Philip McMichael's hypothesis that 'so-called "unlimited supplies of labour" from the countryside metamorphose into a seemingly unlimited supply of unemployed slum-dwellers, exiting increasingly degraded habitats' (McMichael, 2008, p. 219). Similarly, probing hypotheses derived from such

emancipated approaches promise to generate important insight from the already-available aggregated data commonly used by economists. Findings from this investigation have proven productive in this way, namely by demonstrating how local people see how environmental well-being and liberty to act on traditional ecological obligations feed into their will and ability to remain in the cultural landscapes they inherited.

On an entirely distinct note, it must be noted that our investigation elicited the troubling sentiment of disenchantment with locally perceived cultural diversity, a sentiment with an unfortunate degree of emerging potency in global eco-politics (Bacon, 2019; Bhatt, 2020). In two particular sites, participants overtly explained their livelihood struggles by bemoaning how cultural diversity diminished local social solidarity. At the very least, this points to the need for further research into this sensitive factor as it regards local environmental attitudes in general and perceived sovereignty over environmental policy, in particular. Perhaps more importantly, it points to the linkages between peoples' concepts of their belonging in environment and in society and how these might be addressed in tandem from a policy perspective.

Finally, our findings also offer insight for more granular studies of the socio-ecological conditions which precede, underlie and succeed rural abandonment. First, our findings reinforce the growing body of ecological studies from Eurasia which problemetise rural abandonment on the grounds that culturally valuable landscape features are a valid indicator in the determination of ecological health, especially in highly anthropogenic landscapes (Otero et al., 2015; Queiroz et al., 2014). Relatedly, our study further establishes the importance of conducting in-depth human ecological research in contexts where local culturally valuable environmental features are under imminent threat. This model, put forward by the ethnographer Sakakibara (2017) and ethnobiologists Stoffle et al. (1990), demonstrates that in moments of ecological threat, local cultural value for the endangered environmental feature intensifies and redoubles. On the precipice of the sixth mass extinction (Barnosky et al., 2011), further studies are necessary to better understand and support this timely motivating social force for conservation.

Acknowledgments

The authors are highly indebted to the Turkish National General Directorate of Forestry, especially their offices and personnel in Borçka, Bursa, Çanakkale, Rize, Şile, Sinop, Trabzon and Zonguldak, for their invaluable advice and support. We are also very thankful for the administration of Istanbul University Faculty of Forestry for facilitating our research efforts. We are thankful to Yılmaz Erdal, chair of the Anthropology Department at Hacettepe University for his invaluable advising. We sincerely thank The Scientific and Technological Research Council of Turkey (TUBITAK), American Research Institute in Turkey, the U.S. Borlaug Fellows in Global Food Security Program and the Turkish Fulbright Commission for their generous support of this research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Scientific and Technological Research Council of Turkey, the American Research Institute in Turkey, the Turkish Fulbright Commission, and the U.S. Borlaug Fellows Graduate Research Grant under grant [8000075690];

Notes on contributors

Jeffrey Wall is a postdoctoral associate at University of Guelph's Department of Geography, Environment and Geomatics. His research interests include cultural dimensions of biological conservation, ethnobiology and anthropological concepts of value.

Nesibe Köse is a professor at the Istanbul University-Cerrahpaşa, Faculty of Forestry in the Department of Forest Botany. Her interests focus on dendroclimatological reconstructions and dendroecological, dendroarcheological and dendro-geomorphological research using tree rings.

Elif Aksoy is a lecturer at the Hacettepe University Department of Anthropology. Her interests include gender studies and cultural differentiation.

Coşkun Köse is a professor at the Istanbul University-Cerrahpaşa, Faculty of Forestry in the Department of Forest Biology and Wood Protection Technology. His interests include wood microbiology, conservation of wood artefacts, and non-wood forest products industry.

Taner Okan is an associate professor at the Istanbul University-Cerrahpaşa, Faculty of Forestry in the Department of Forestry Economics. His research interests include non-timber forest product development, value chains, and forestry conflict resolution.

Shorna Allred is an associate professor at Cornell University's Department of Natural Resources. Her research has focused on the decision-making and motivations behind individual biological conservation.

ORCID

Jeffrey Robert Wall http://orcid.org/0000-0002-2775-8132
Nesibe Köse http://orcid.org/0000-0001-5766-0526
Elif Başak Aksoy http://orcid.org/0000-0002-7709-8582
Coşkun Köse http://orcid.org/0000-0003-4626-2236
Taner Okan http://orcid.org/0000-0001-7531-5662
Shorna Allred http://orcid.org/0000-0001-6237-0638

References

Akdogan, S., and Erkman, E. (1968). Dikkat kestane kanseri görüldü. *Tomurcuk*, *1*, 4–5

Akilli, S., Katircioglu, Y. Z., & Maden, S. (2009). Vegetative compatibility types of Cryphonectria parasitica, causal agent of chestnut blight, in the Black Sea region of Turkey. *Forest Pathology*, *39*(6), 390–396. https://doi.org/10.1111/j.1439-0329.2009.00601.x

Aumeeruddy-Thomas, Y., Therville, C., Lemarchand, C., Lauriac, A., & Richard, F. (2012). Resilience of sweet chestnut and Truffle Holm-Oak rural forests in Languedoc-Roussillon, France : Roles of social-ecological legacies, domestication, and innovations. *Ecology and Society*, *17*(2), 12–24. https://doi.org/10.5751/ES-04750-170212

Bacon, J. (2019). Settler colonialism as eco-social structure and the production of colonial ecological violence. *Environmental Sociology*, *5*(1), 59–69. https://doi.org/10.1080/23251042.2018.1474725

Barnosky, A. D., Matzke, N., Tomiya, S., Wogan, G. O. U., Swartz, B., Quental, T. B., Marshall, C., McGuire, J. L., Lindsey, E. L., Maguire, K. C., Mersey, B., & Ferrer, E. A. (2011). Has the Earth's sixth mass extinction already arrived? *Nature*, *471* (7336), 51–57. https://doi.org/10.1038/nature09678

Berkes, F., Colding, J., & Folke, C. (2000). Rediscovery of traditional ecological knowledge as adaptive management. *Ecological Applications*, *10*(5), 1251–1262. https://doi.org/10.1890/1051-0761(2000)010[1251:ROTEKA]2.0.CO;2

Bhatt, C. (2020). White extinction: Metaphysical elements of contemporary Western fascism. *Theory, Culture and Society*, *38*(1), 27–52. https://doi.org/10.1177/0263276420925523

Bugday, S. E., & Özden, S. (2017). The relationship between terrain and rural migration (1965 – 2013) on the north of Turkey (the case of Kastamonu. *Environmental Monitoring and Assessment*, *189*(154), 1–11. https://doi.org/10.1007/s10661-017-5867-9

Cetin, G., Orman, E., and Polat, Z. (2014). First record of the Oriental gall wasp, Dryocosmus kuriphilus Yasumatsu (Hymenoptera: Cynipidae) in Turkey. Bitki Koruma Bulteni, 54, 303–309

Davis, M. (2006). *Planet of Slums*. Verso.

Erdem, R. (1951). Causes of Die-Back of Chestnut Trees in Turkey and Control of Disease. *Tarım Bakanlığı, Orman Genel Müdürlüğü*, Sayı No: 102, Seri 11, Ankara.

FAO. (2009). *Report on chestnut blight and its management in Turkey*. Ankara.

Ghelardini, L., Pepori, A. L., Luchi, N., Capretti, P., & Santini, A. (2016). Drivers of emerging fungal diseases of forest trees. *Forest Ecology and Management*, *381*, 235–246. https://doi.org/10.1016/j.foreco.2016.09.032

Hobbs, R. J., & Cramer, V. (2007). Why old fields?: Socioeconomic and ecological causes of land abandonment. In R. J. Hobbs (Ed.), *Old fields: Dynamics and restoratoin of abandoned farmland,* 1–15. Island Press.

Lewis, W. A. (1954). Economic Development with Unlimited Supplies of Labour. *Manchester School of Economic and Social Studies*, *22*, 139–191.

Liao, P.-J., Wang, P., Wang, Y.-C., & Chong, K. Y. (2017). *Education Choice, Rural-urban Migration and Economic Development*. DOI 10.3386/w23939

Longstaff, P. H., Armstrong, N. J., Perrin, K., May, W., & Hidek, M. A. (2010). Building resilient communities : A preliminary framework for assessment. *Homeland Security Affairs*, *6*(3), 1–23. https://securitypolicylaw.syr.edu/wp-content/uploads/2012/09/Building-Resilient-Communities.pdf

Luken, J. O. (1990). *Directing Ecological Succession*. Chapman and Hall.

Mather, A. S. (1992). The Forest Transition. *Area*, *24*(4), 367–379. https://www.jstor.org/stable/20003181

Mattioni, C., Martin, M. A., Cherubini, M., Taurchini, D., & Villani, F. (2010). Genetic diversity in European chestnut populations. *Actahort*, *866*, 163–168. https://doi.org/10.17660/ActaHortic.2010.866.18

McMichael, P. (2008). Peasants make their own history, but not just as they please. *Journal of Agrarian Change*, *8*(2), 205–228. https://doi.org/10.1111/j.1471-0366.2008.00168.x

Michon, G. (2011). Revisiting the resilience of chestnut forests in Corsica : From social-ecological sytems theory to political ecology. *Ecology and Society*, *16*(2), 4–16. https://www.jstor.org/stable/26268881

Miles, M., & Huberman, A. (1984). *Qualitative data analysis: A sourcebook of new methods*. Sage Publications.

Navarro, L. M., & Pereira, H. M. (2015). Rewilding abandoned landscapes in Europe. *Rewilding European Landscapes*, 3–23. Springer: New York. https://doi.org/10.1007/978-3-319-12039-3_1

Okan, T., Kose, C., Aksoy, E., Kose, N., & Wall, J. (2017). Turkiye'de Kestane e (*Castanea sativa* Mill.) ve Kullanimi Uzerine Geleneksel Terimler. *Avrasya Terim Dergisi*, *5*(1), 19–27.

Okan, T., Köse, C., Köse, N., Aksoy, E. B., & Wall, J. R. (2021). How managing for chestnut honey in Turkey salvages trees and lifeways under increasing exotic pest and disease pressure. *Human Ecology*, *49*(2), 205–216. https://doi.org/10.1007/s10745-021-00220-5

Otero, I., Marull, J., Tello, E., Diana, G. L., Pons, M., Coll, F., & Boada, M. (2015). Land abandonment, landscape, and biodiversity: Questioning the restorative character of the forest transition in the Mediterranean. *Ecology and Society*, *20*(2), 2. https://doi.org/10.5751/ES-07378-200207

Parry, L., Day, B., Amaral, S., & Peres, C. A. (2010). Drivers of rural exodus from Amazonian headwaters. *Population and Environment*, *32*(2–3), 137–176. https://doi.org/10.1007/s11111-010-0127-8

Pezzi, G., Lucchi, E., Maresi, G., Ferretti, F., Viaggi, D., & Frascaroli, F. (2017). Land use policy abandonment or survival? Understanding the future of Castanea sativa stands in function of local attitude (Northern Apennine, Italy). *Land Use Policy*, *61*, 564–574. https://doi.org/10.1016/j.landusepol.2016.10.049

Plieninger, T., Hui, C., Gaertner, M., Huntsinger, L., & Webb, E. (2014). The impact of land abandonment on species richness and abundance in the Mediterranean Basin: A meta-analysis. *PLoS ONE*, *9*(5), 5. https://doi.org/10.1371/journal.pone.0098355

Queiroz, C., Beilin, R., Folke, C., & Lindborg, R. (2014). Farmland abandonment: Threat or opportunity for biodiversity conservation? A global review. *Frontiers in Ecology and the Environment*, *12*(5), 288–296. https://doi.org/10.1890/120348

Rapport, D., & Maffi, L. (2010). The dual erosion of biological and cultural diversity: Implications for the health of ecocultural systems. In S. Pilgrim & J. Pretty (Eds.), *Nature and culture: Rebuilding lost connections* (pp. 103–122). Earthscan.

Rostow, W. (1960). *The Stages of Economic Growth: A Non-Communist Manifesto*. Cambridge University Press.

Sakakibara, C. (2017). People of the whales: Climate Change and cultural resilience among Iñupiat of Arctic Alaska. *Geographical Review*, *107*(1), 159–184. https://doi.org/10.1111/j.1931-0846.2016.12219.x

Sanz, A., Fernandez, C., Mouillot, F., Istria, D., & Pasqualini, V. (2013). Long-term forest dynamics and land-use abandonment in the Mediterranean Mountains, Corsica, France. *Ecology and Society*, *18*(2), 37–53. https://www.jstor.org/stable/26269315

Stoffle, R., Halmo, D., Evans, M., & Olmsted, J. (1990). Calculating the cultural significance of American Indian plants: Paiute and shoshone ethnobotany at Yucca Mountain, Nevada. *American Anthropologist1*, *92*(2), 416–432. https://doi.org/10.1525/aa.1990.92.2.02a00100

TNGDF. (2013). *Kestane Eylem Plani 2013-2017*. Orman ve Su İşleri Bakanlığı, Orman Genel Müdürlüğü Yayınları

Topcu, K., Serdar, U., Ozturk, A., & Genc, M. (2009). A chestnut monument tree in Sinop: Padisah Kestanesi. *Acta Horticulturae*, *815*, 163–170. https://doi.org/10.17660/ActaHortic.2009.815.21

Voeks, R. A. (2007). Are women reservoirs of traditional plant knowledge? Gender, ethnobotany and globalization in northeast Brazil. *Singapore Journal of Tropical Geography*, *28*(1), 7–20. https://doi.org/10.1111/j.1467-9493.2006.00273.x

Wall, J., Köse, C., Köse, N., Okan, T., Aksoy, E., Jarvis, D., & Allred, S. (2019). The role of traditional livelihood practices and local ethnobotanical knowledge in mitigating chestnut disease and pest severity in Turkey. *Forests*, *10*(7), 571. https://doi.org/10.3390/f10070571

Warner, M. (2002). *Publics and Counterpublics*. Zone Books.

Zlatanov, T., Schleppi, P., Velichkov, I., Hinkov, G., Georgieva, M., Eggertsson, O., Zlatanova, M., & Vacik, H. (2013). Structural diversity of abandoned chestnut (Castanea sativa Mill .) dominated forests : Implications for forest management Forest Ecology and Management Structural diversity of abandoned chestnut (Castanea sativa Mill .) dominated forests : Implications. *Forest Ecology and Management*, *291*, 326–335. https://doi.org/10.1016/j.foreco.2012.11.015

View publication stats